Python

POLLUTION AND HEALTH

Group 11

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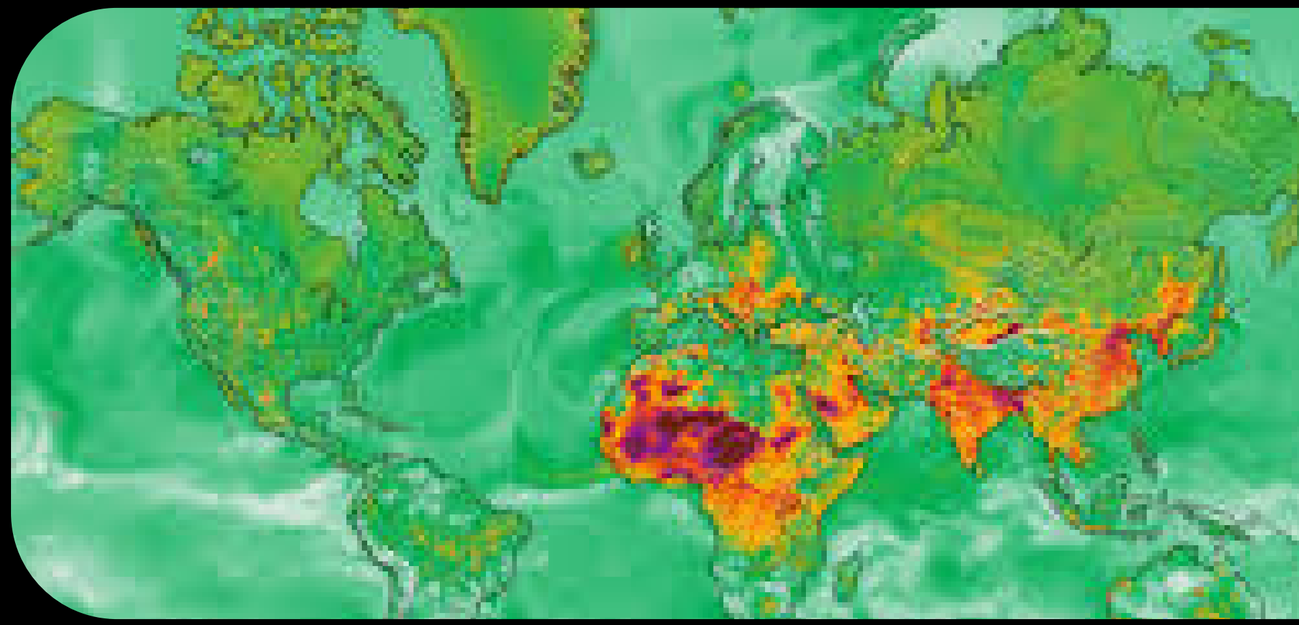
Core **Idea**

WHY AQI?

Importance of AQI data

Finding Correlation between city proplation, airquality and health impacts are first steps to identifying key causes of different issues both health and others

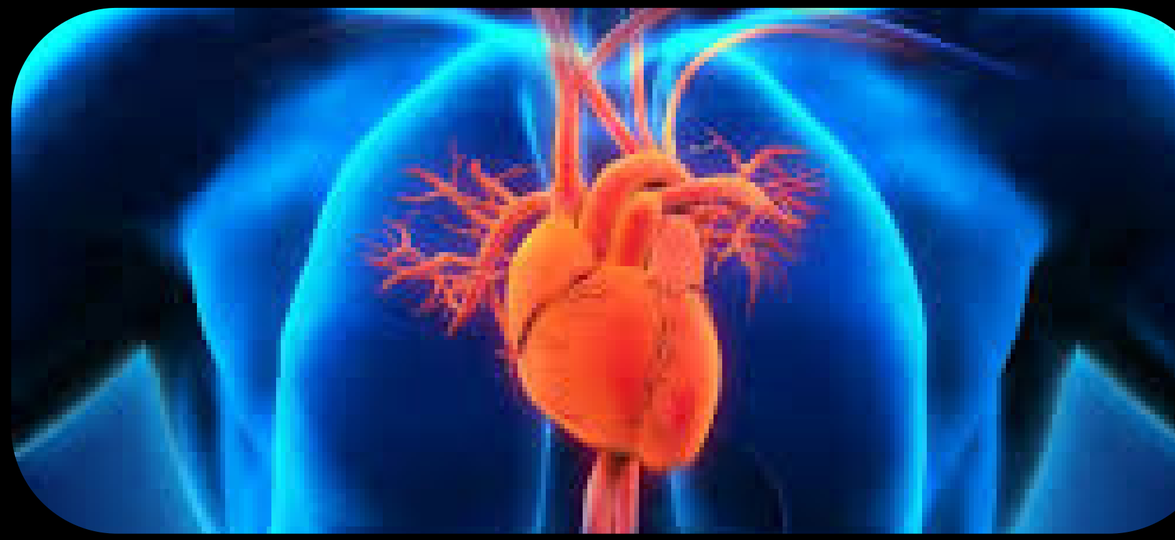
DATASET #1
AQI DATA



Air quality index

AQI is measured by monitoring concentrations of key pollutants—primarily fine particulate matter **PM2.5**, **PM10**, ozone **O3**, nitrogen dioxide **NO2**, sulfur dioxide **SO2**, and carbon monoxide **CO**

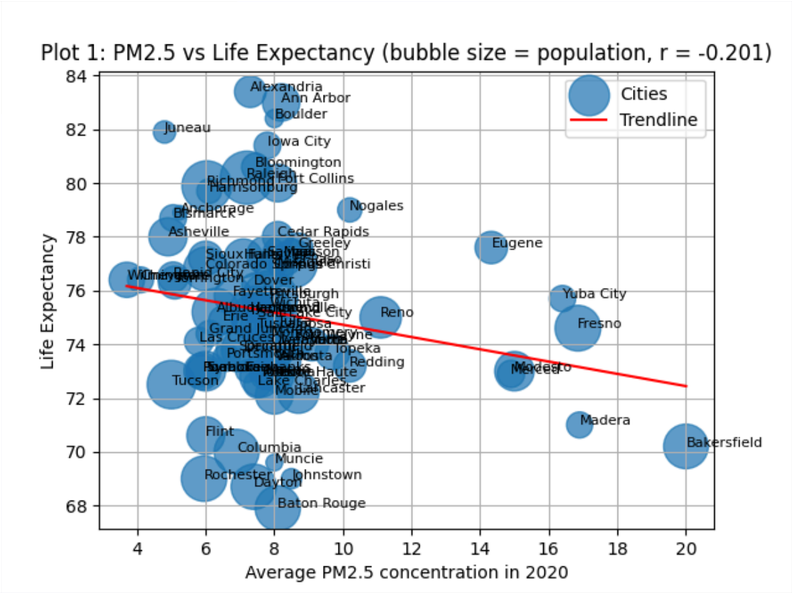
DATASET #2
HEALTH STATISTICS

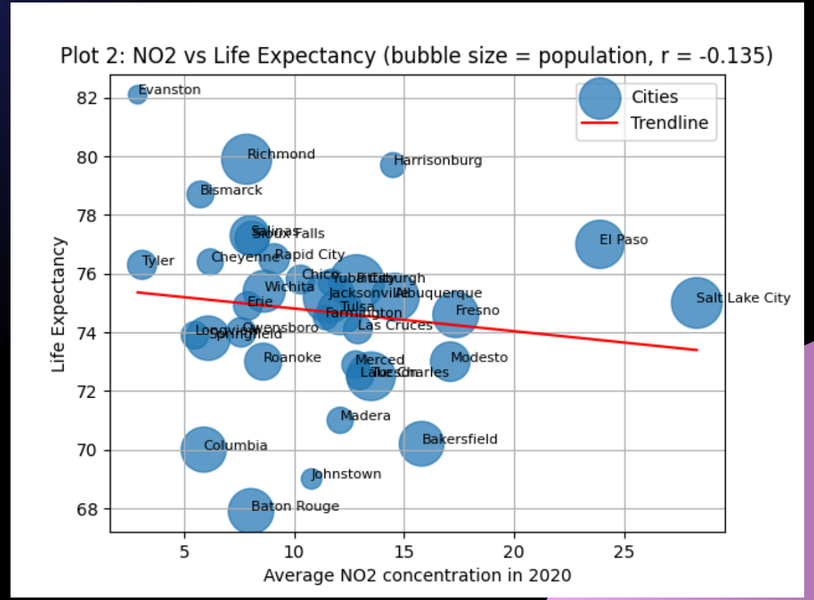


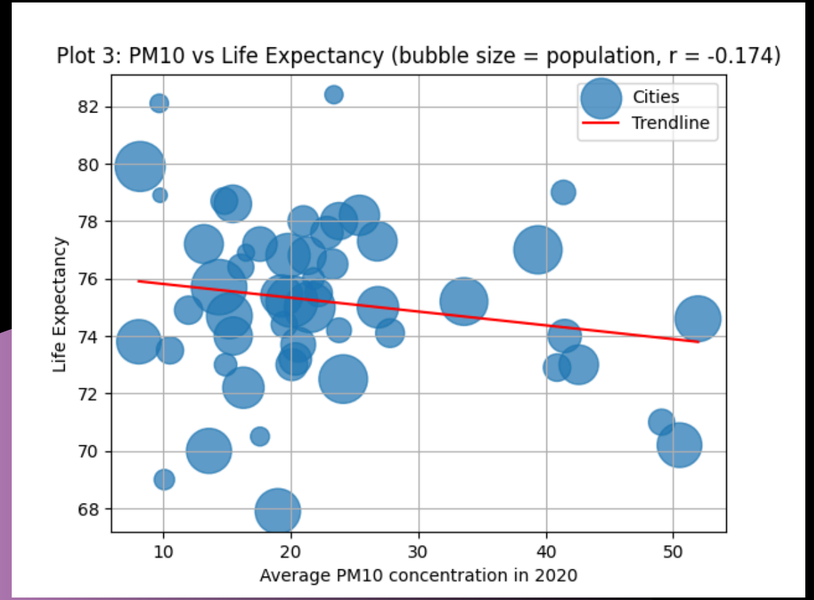
Population data from **1000+** cities

Statistics of **Health Indicators** like cardiovascular disease or even death

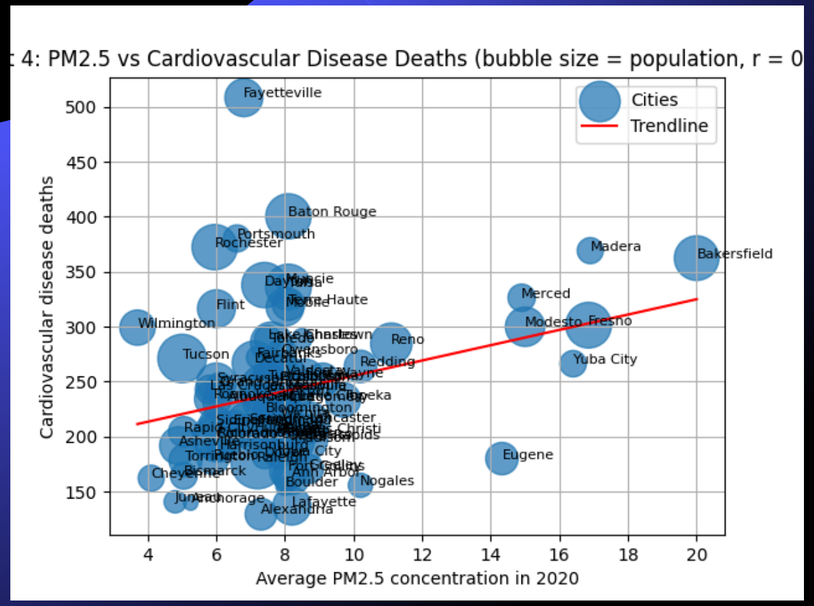
DATA PLOTS

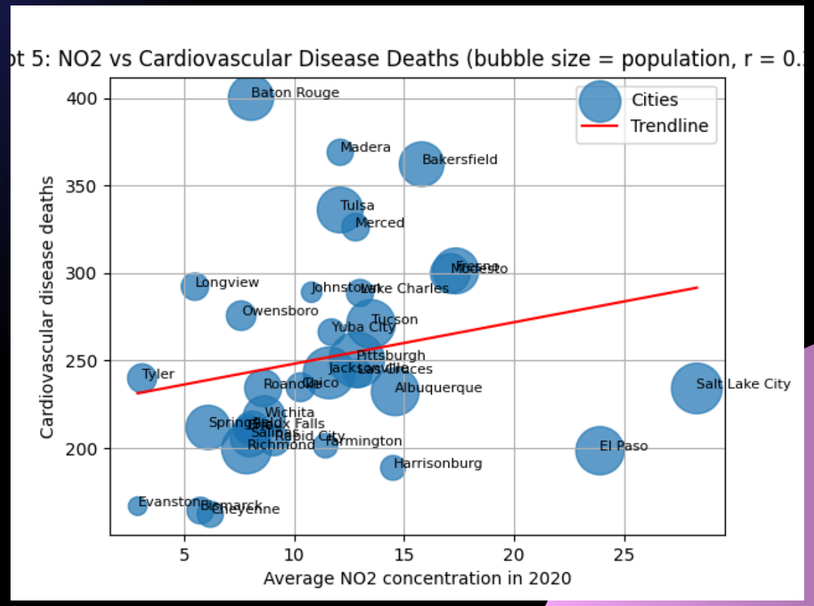

Plot 1: PM2.5 vs Life Expectancy (bubble size = population, r = -0.201)
Cities
Trendline
Life Expectancy
Average PM2.5 concentration in 2020
Alexandria
Ann Arbor
Boulder
Juneau
Iowa City
Bloomington
Nogales
Asheville
Cedar Rapids
Eugene
Yuba City
Reno
Fresno
Madera
Bakersfield
Tucson
Flint
Columbia
Muncie
Rochester
Johnstown
Dayton
Baton Rouge

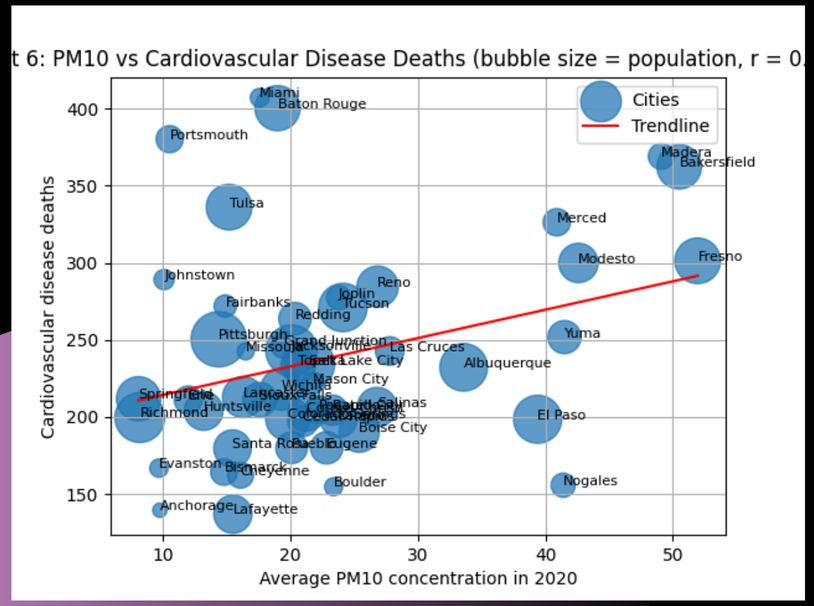

Plot 2: NO2 vs Life Expectancy (bubble size = population, r = -0.135)
Cities
Trendline
Life Expectancy
Average NO2 concentration in 2020
Evanston
Richmond
Harrisonburg
Bismarck
El Paso
Salt Lake City
Tyler
Cheyenne
Rapid City
Wichita
Erie
Fresno
Las Cruces
Roanoke
Modesto
Madera
Bakersfield
Columbia
Johnstown
Baton Rouge


Plot 3: PM10 vs Life Expectancy (bubble size = population, r = -0.174)
Cities
Trendline
Life Expectancy
Average PM10 concentration in 2020

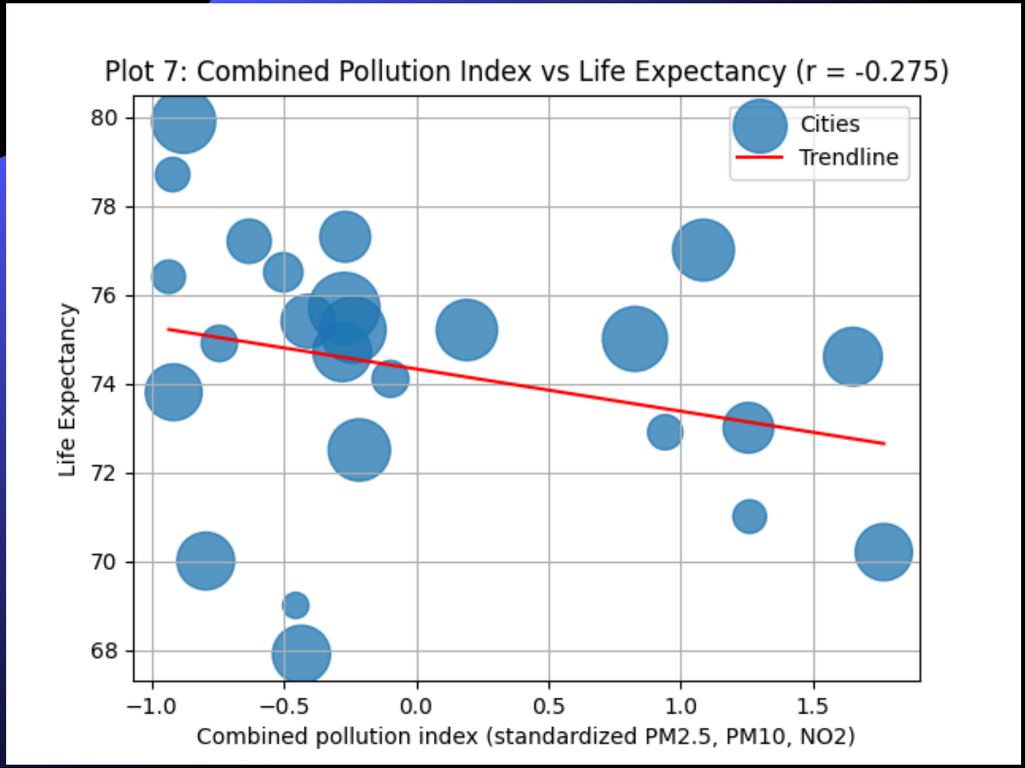
DATA PLOTS

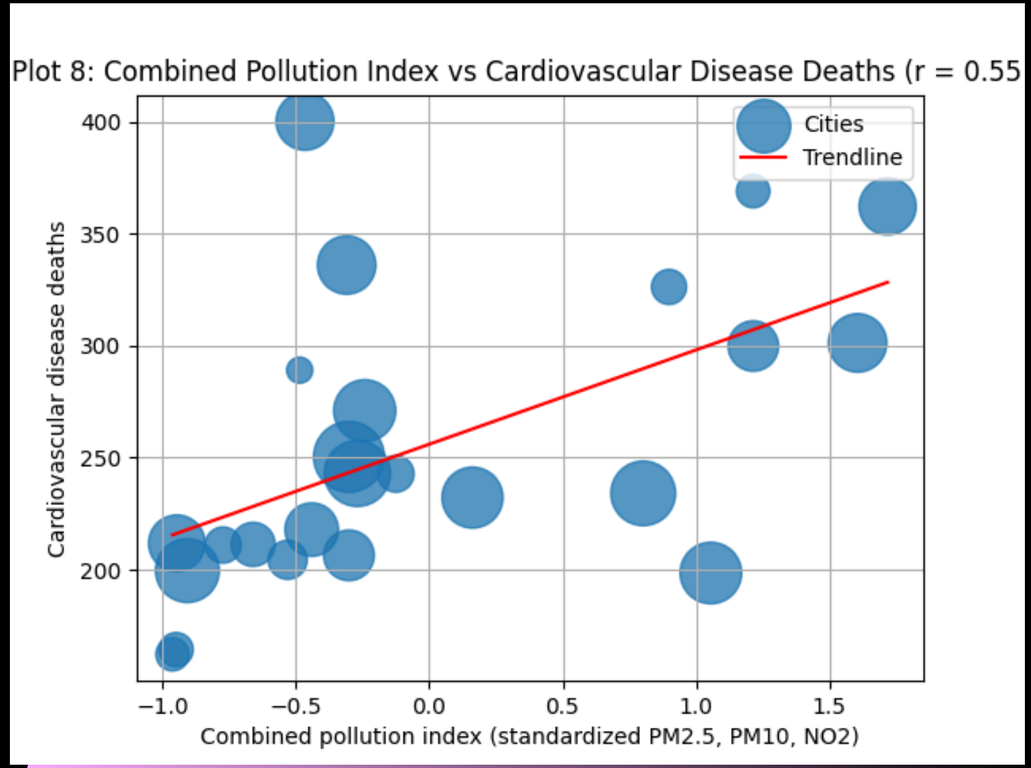

t 4: PM2.5 vs Cardiovascular Disease Deaths (bubble size = population, r = 0
Cities
Trendline
Cardiovascular disease deaths
Average PM2.5 concentration in 2020
Fayetteville
Baton Rouge
Portsmouth
Rochester
Madera
Bakersfield
Merced
Modesto
Fresno
Yuba City
Wilmington
Tucson
Reno
Redding
Eugene
Nogales
Boulder
Lafayette
Alexandria
Anchorage
Cheyenne
Bismarck
Flint
Terre Haute
Asheville
Torrington
Ann Arbor
Fort Collins


ot 5: NO2 vs Cardiovascular Disease Deaths (bubble size = population, r = 0.
Cities
Trendline
Cardiovascular disease deaths
Average NO2 concentration in 2020
Baton Rouge
Madera
Bakersfield
Tulsa
Merced
Modesto
Fresno
Longview
Johnstown
Lake Charles
Owensboro
Tucson
Yuba City
Pittsburgh
Tyler
Roanoke
Waco
Jacksonville
Albuquerque
Salt Lake City
Wichita
Springfield
Falls City
Richmond
Farmington
El Paso
Harrisonburg
Evanston
Bismarck
Cheyenne

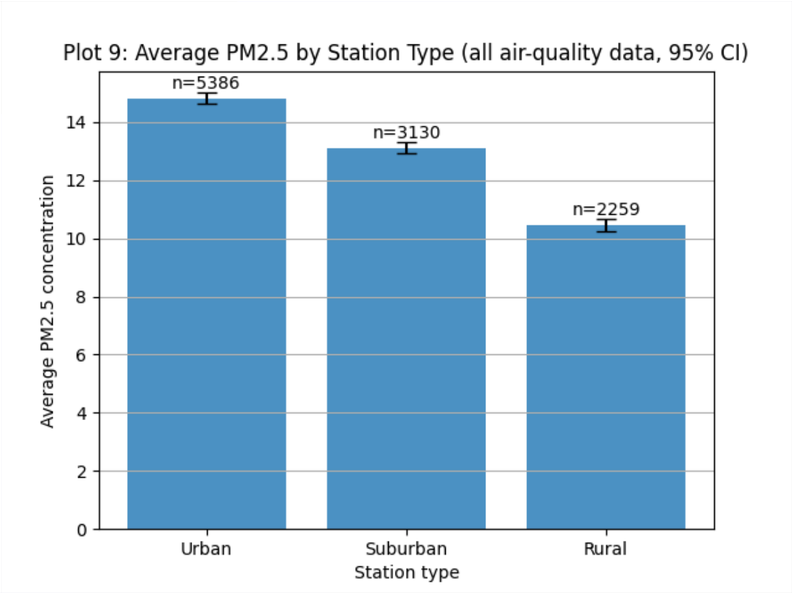

t 6: PM10 vs Cardiovascular Disease Deaths (bubble size = population, r = 0.
Cities
Trendline
Cardiovascular disease deaths
Average PM10 concentration in 2020
Miami
Baton Rouge
Portsmouth
Madera
Bakersfield
Tulsa
Merced
Modesto
Fresno
Johnstown
Reno
Joplin
Tucson
Fairbanks
Redding
Pittsburgh
Yuma
Las Cruces
Salt Lake City
Albuquerque
Mason City
Wichita
Lancaster
Falls
Salinas
Huntsville
Richmond
El Paso
Boise City
Santa Rosa
Eugene
Evanston
Bismarck
Cheyenne
Boulder
Nogales
Anchorage
Lafayette

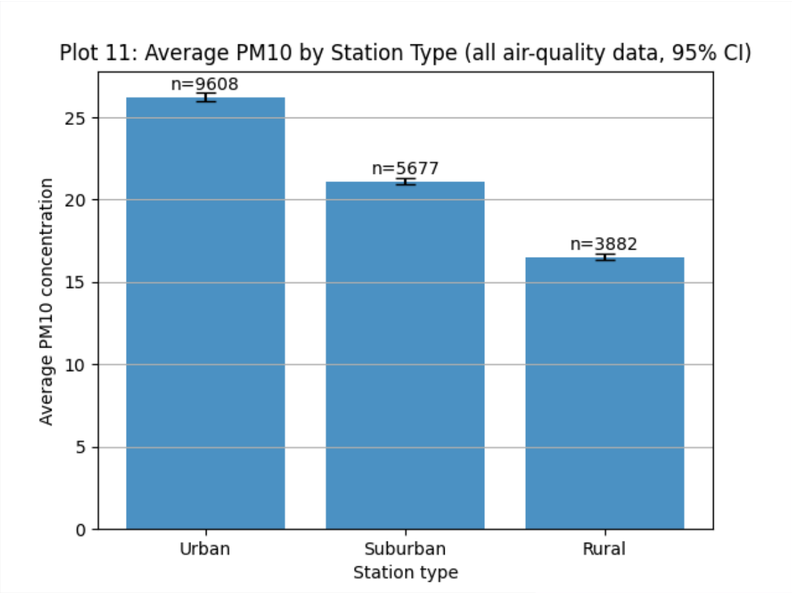
DATA PLOTS

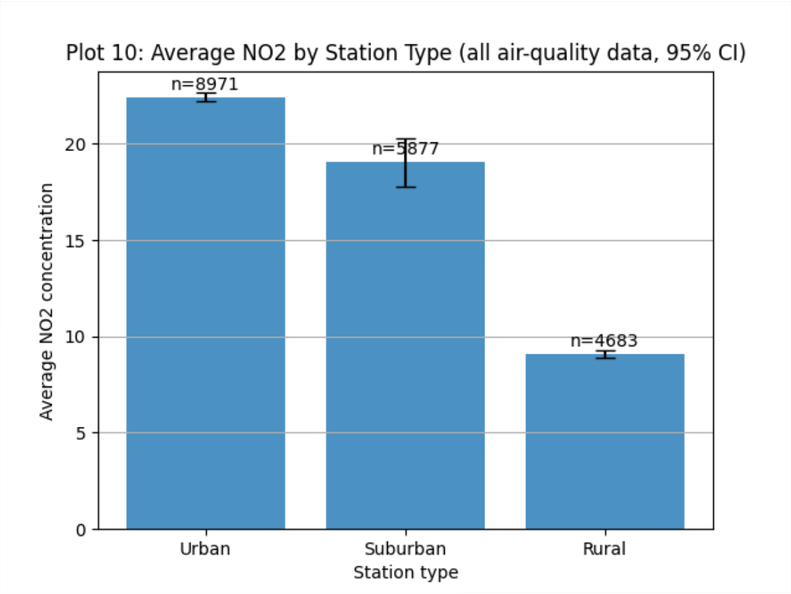

Plot 7: Combined Pollution Index vs Life Expectancy (r = -0.275)
Cities
Trendline
Life Expectancy
80
78
76
74
72
70
68
−1.0
−0.5
0.0
0.5
1.0
1.5
Combined pollution index (standardized PM2.5, PM10, NO2)


Plot 8: Combined Pollution Index vs Cardiovascular Disease Deaths (r = 0.55
Cities
Trendline
Cardiovascular disease deaths
400
350
300
250
200
−1.0
−0.5
0.0
0.5
1.0
1.5
Combined pollution index (standardized PM2.5, PM10, NO2)

DATA PLOTS


Plot 9: Average PM2.5 by Station Type (all air-quality data, 95% CI)
n=5386
n=3130
n=2259
Average PM2.5 concentration
0
2
4
6
8
10
12
14
Urban
Suburban
Rural
Station type


Plot 11: Average PM10 by Station Type (all air-quality data, 95% CI)
n=9608
n=5677
n=3882
Average PM10 concentration
0
5
10
15
20
25
Urban
Suburban
Rural
Station type


Plot 10: Average NO2 by Station Type (all air-quality data, 95% CI)
n=8971
n=5877
n=4683
Average NO2 concentration
0
5
10
15
20
Urban
Suburban
Rural
Station type

```
1 PM 2.5 against Life Expectancy
=analysis_df[["pm25_mean","life_expectancy","population","city_clean"]].dropna(

_df_1["pm25_mean"]
_df_1["life_expectancy"]
rson_r(x1,y1)
np.sqrt(plot_df_1["population"])/np.sqrt(plot_df_1["population"]).max()*1000+30

gure()
atter(x1,y1,s=sizes1,alpha=0.7,label="Cities")
endline(x1,y1)
te_all_points(plot_df_1,"pm25_mean","life_expectancy","city_clean")

tle(f"Plot 1: PM2.5 vs Life Expectancy (bubble size = population, r = {r1:.3f})")
abel("Average PM2.5 concentration in 2020")
abel("Life Expectancy")
gend()
id()
```

REAL WORLD DATA PROCCESING

- IMPORTING
- CLEANING
- MERGING
- PLOTTING

INSIGHTS
FROM THE PROJECT

CONCLUSION

- INCREASE IN POLLUTION SHOWS POSITIVE CORELATION WITH DEATH
- INCREASE IN POLLUTION ALSO SHOWS NEGATIVE CORELATION WITH
- POLUTION IS HIGHER IN URBAN POPULATION AREAS

Thank **You**